

Solution Requirements

Date	15 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Secure login using Google OAuth/email and password with MFA support.	High
2	Authorization levels	Admin, Teacher, Student roles with role-based access.	High
3	External interfaces	Google Cloud Vertex AI, Speech-to-Text APIs, Firebase integration.	High

4	Transactions processing	Process emotion analysis and learning recommendations securely.	High
5	Reporting	Generate emotion analytics dashboards and reports.	Medium
6	Business rules, etc.	Personalized learning recommendations based on detected emotions.	High
7	Compliance to laws or regulations	Comply with GDPR and privacy regulations.	High
8	<i>Other</i>	Real-time alerts for prolonged negative emotions.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	System responds within 2 seconds.	< 2 seconds
2	Scalability	Supports up to 10,000 users.	10,000 concurrent users
3	Security & Data Privacy	Encrypt data using AES-256 and HTTPS.	AES-256
4	Reliability & Availability	Highly available cloud deployment.	99.9% uptime
5	Usability & Accessibility	Accessible, user-friendly interface.	WCAG 2.1 AA
6	<i>Other</i>	Modular and portable architecture.	Easy maintenance